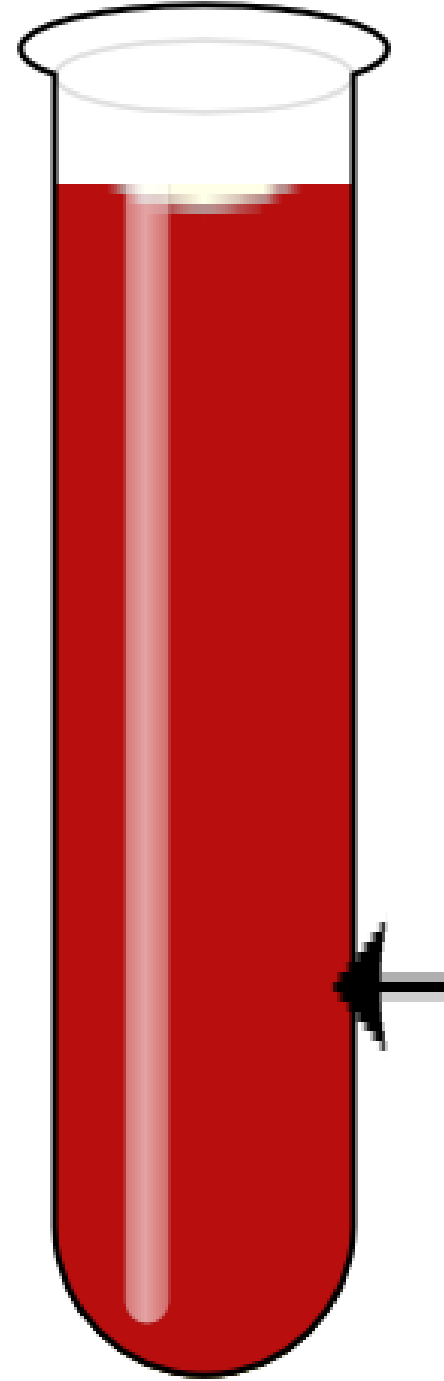


Blood Composition

Blood Composition

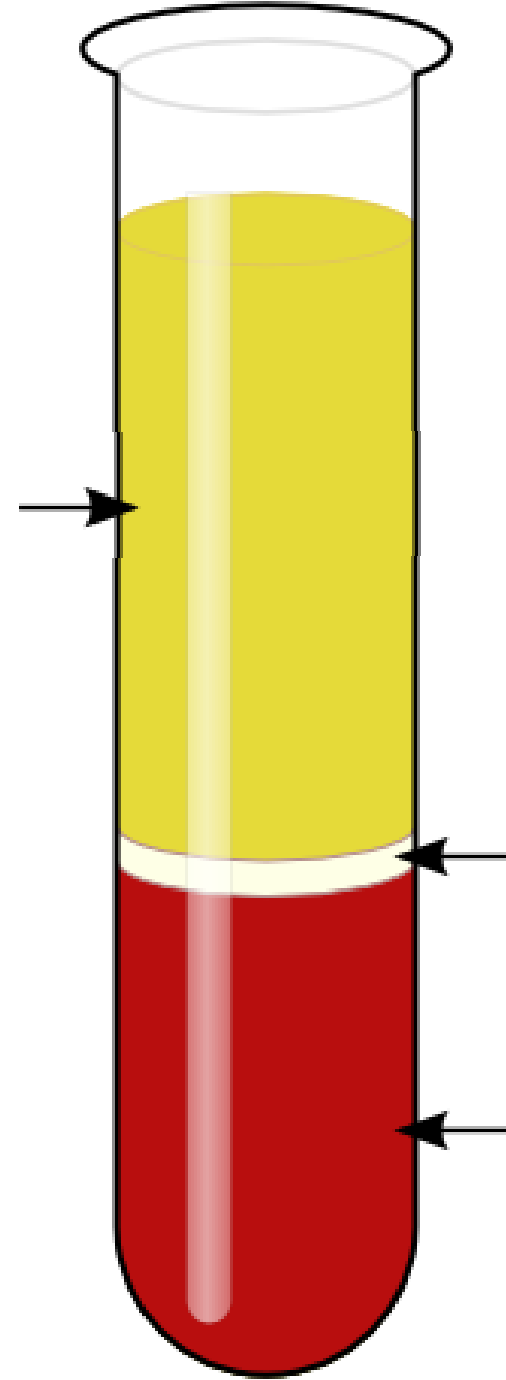


Blood Composition

~55% = plasma

- Coagulation proteins
- Other proteins/molecules

Plasma vs Serum?



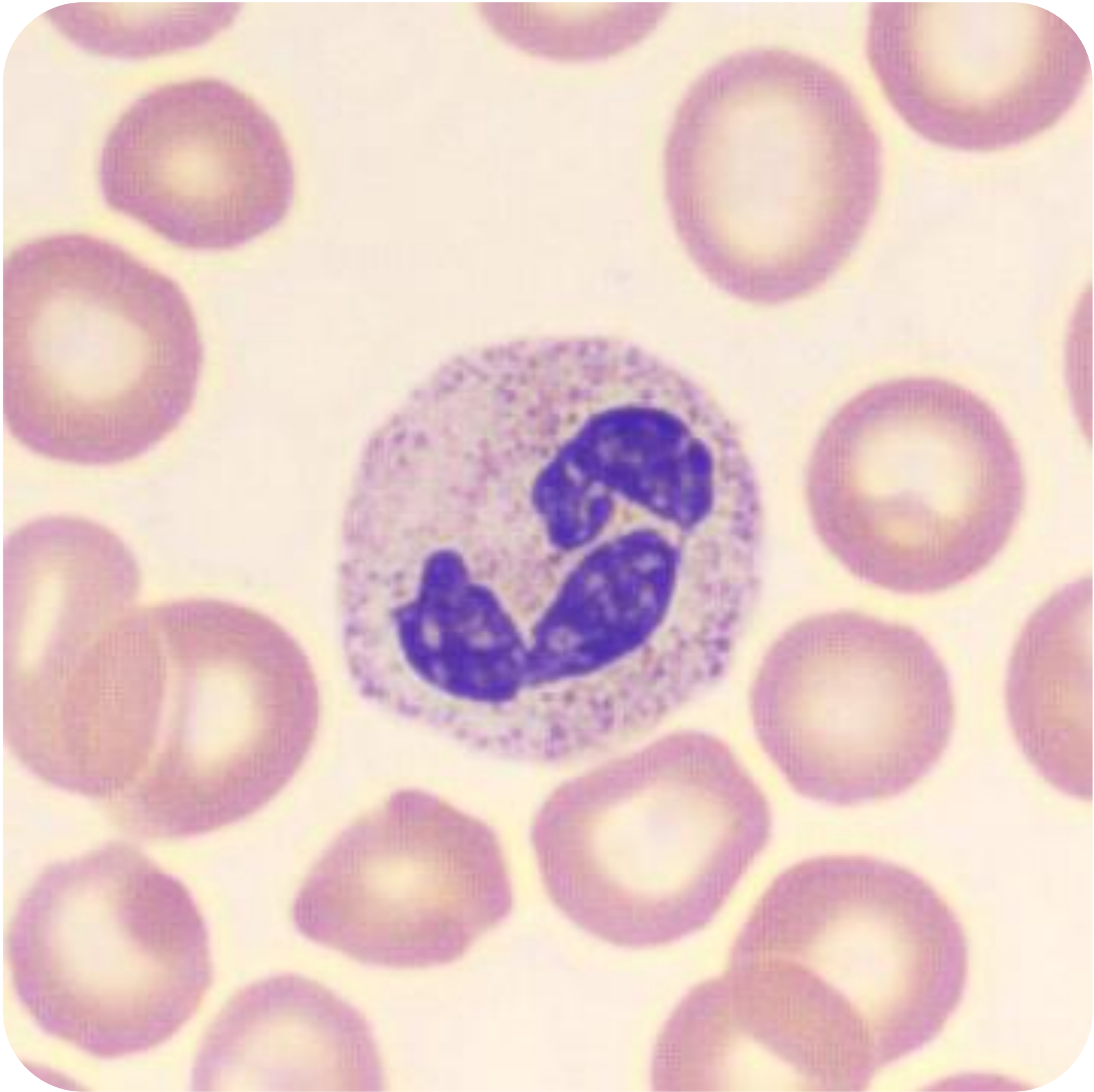
~45% = cellular elements

- WBCs = leukocytes: foreign substance defense
 - Neutrophils; Lymphocytes; Monocytes; Eosinophils; Basophils
- PLTs = thrombocytes (hemostasis)
- RBCs = erythrocytes: transport O_2 and CO_2

WBC - Segmented Neutrophil

- Role?
- Reference Range?

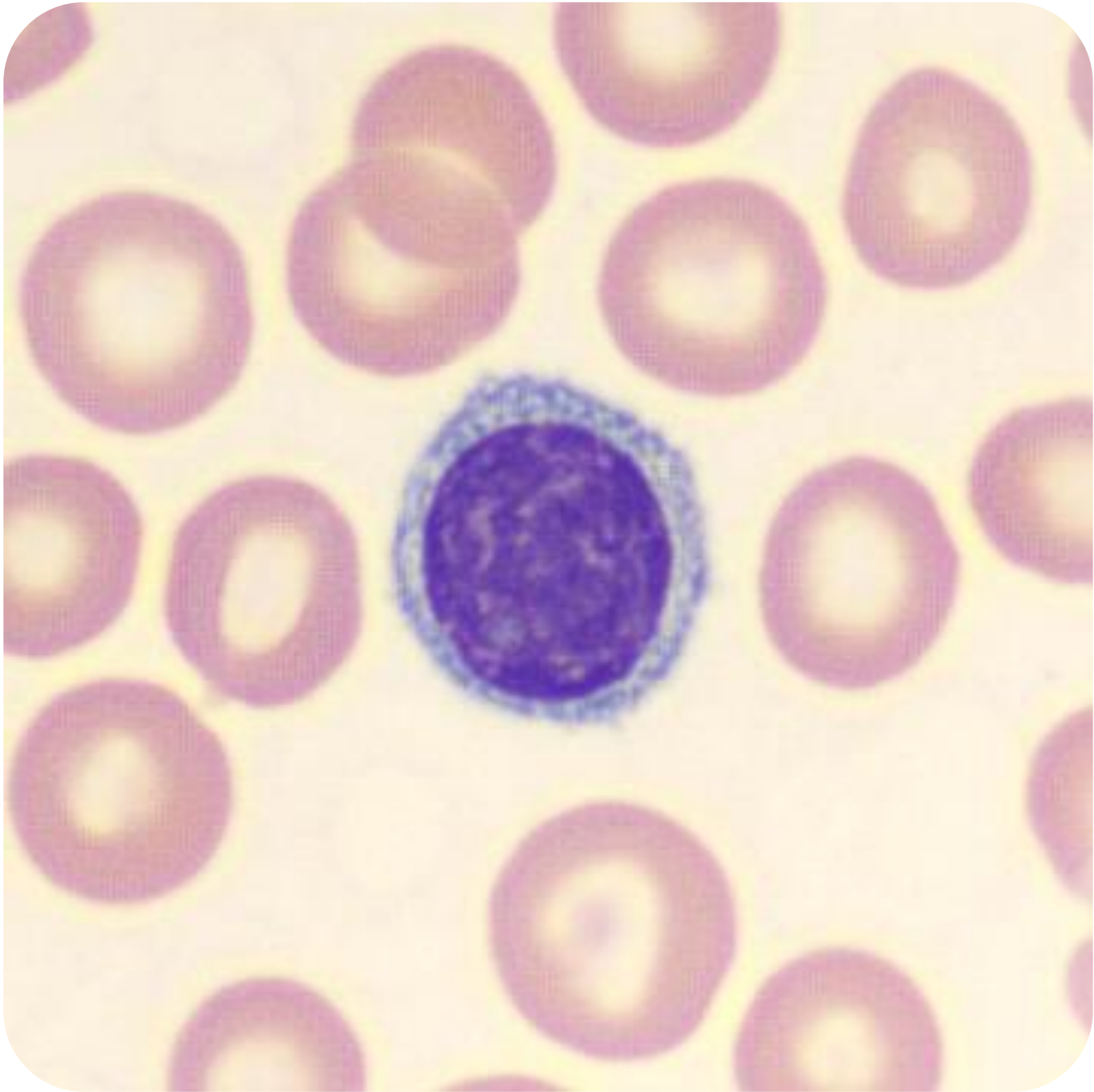
	Conventional Units	SI Units
WBC Total	$3.6 - 10.6 \times 10^3/\mu\text{L}$	$3.6 - 10.6 \times 10^9/\text{L}$
NE, Relative	50 – 70%	0.50 – 0.70
NE, Absolute	$1.7 - 7.5 \times 10^3/\mu\text{L}$	$1.7 - 7.5 \times 10^9/\text{L}$



WBC - Lymphocyte

- Role?
- Reference Range?

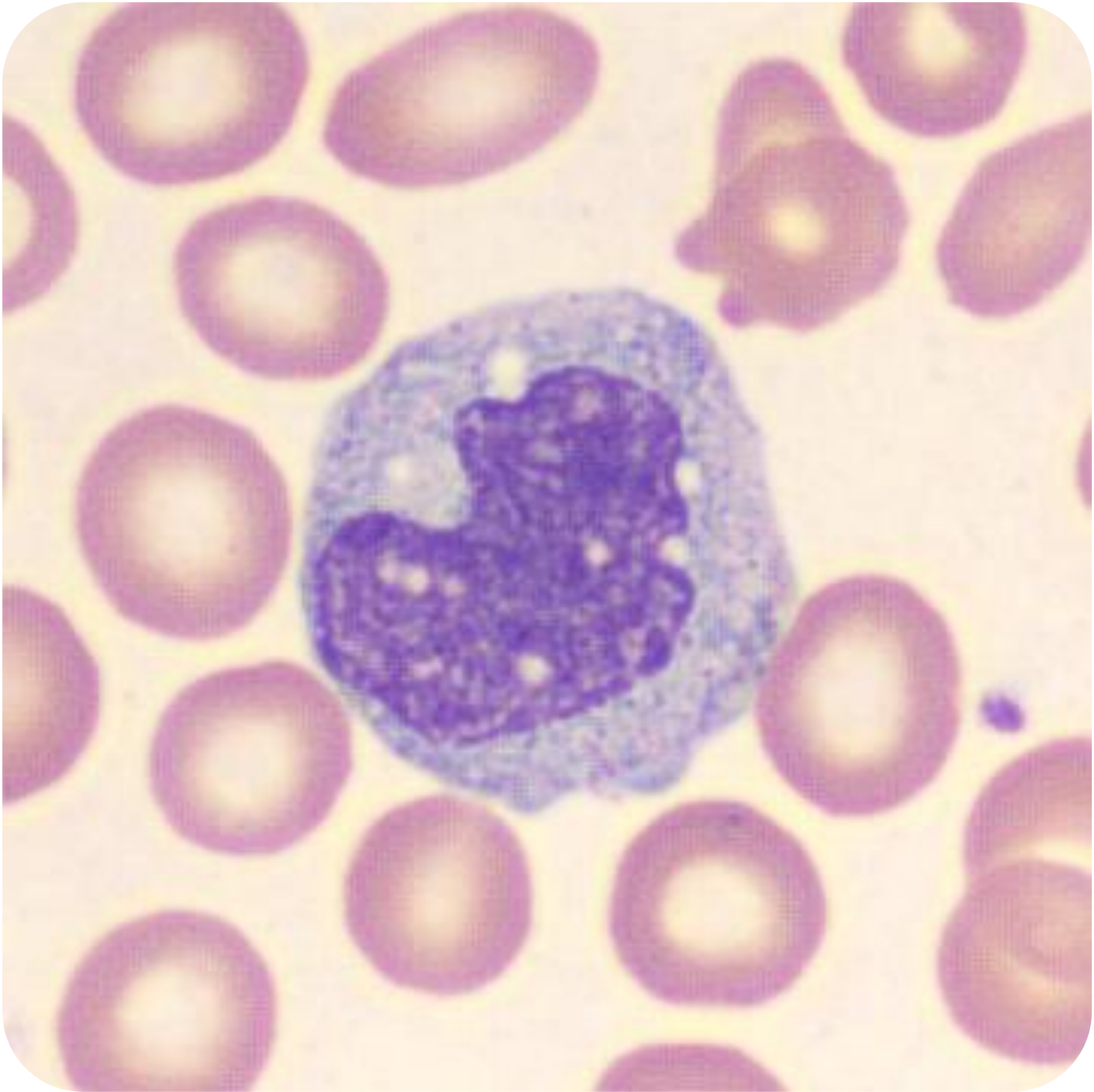
	Conventional Units	SI Units
WBC Total	$3.6 - 10.6 \times 10^3/\mu\text{L}$	$3.6 - 10.6 \times 10^9/\text{L}$
LY, Relative	18 – 42%	0.18 – 0.42
LY, Absolute	$1.0 - 3.2 \times 10^3/\mu\text{L}$	$1.0 - 3.2 \times 10^9/\text{L}$



WBC - Monocyte

- Role?
- Reference Range?

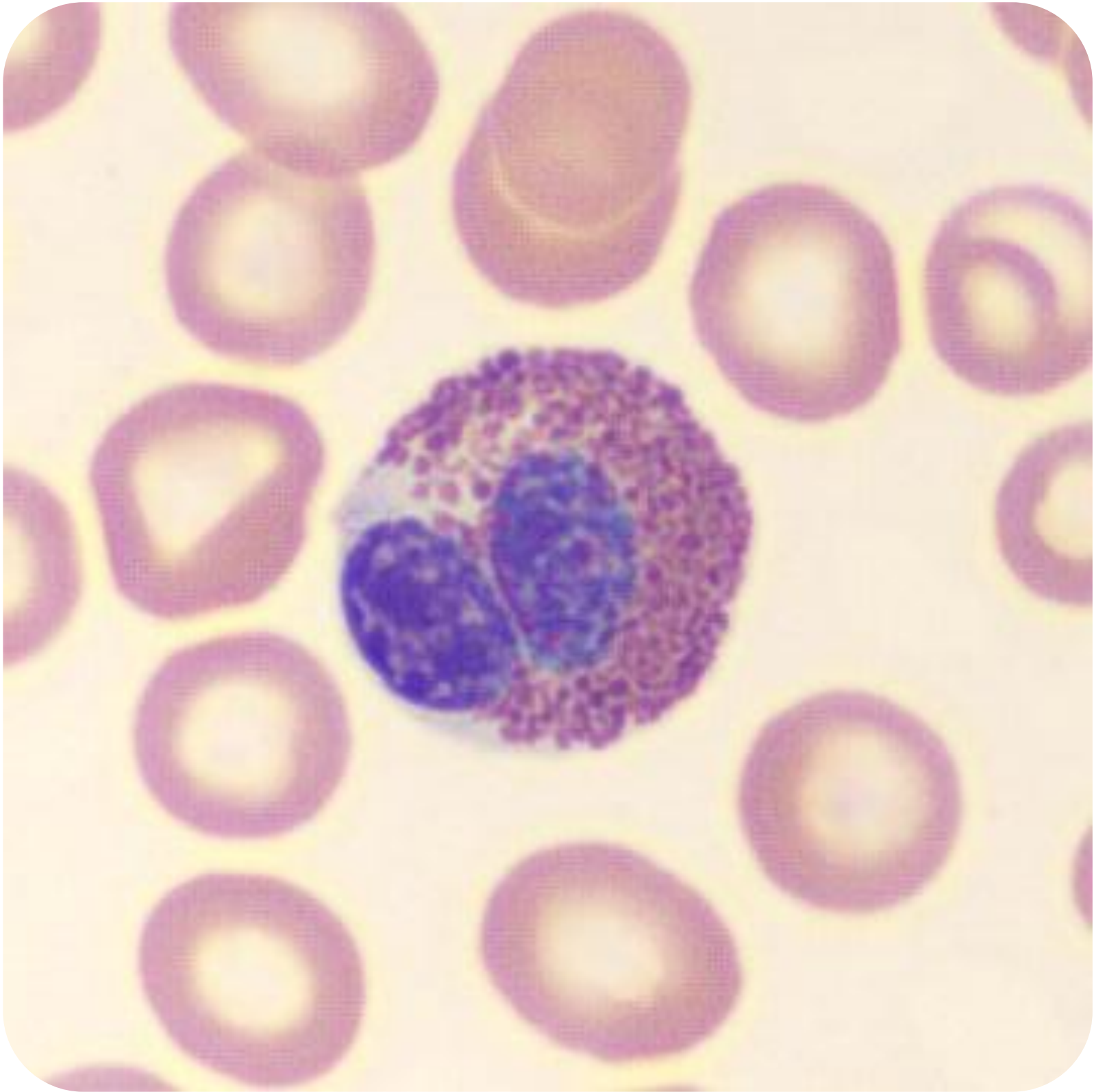
	Conventional Units	SI Units
WBC Total	$3.6 - 10.6 \times 10^3/\mu\text{L}$	$3.6 - 10.6 \times 10^9/\text{L}$
MO, Relative	2 – 11%	0.02 – 0.11
MO, Absolute	$0.1 - 1.3 \times 10^3/\mu\text{L}$	$0.1 - 1.3 \times 10^9/\text{L}$



WBC - Eosinophil

- Role?
- Reference Range?

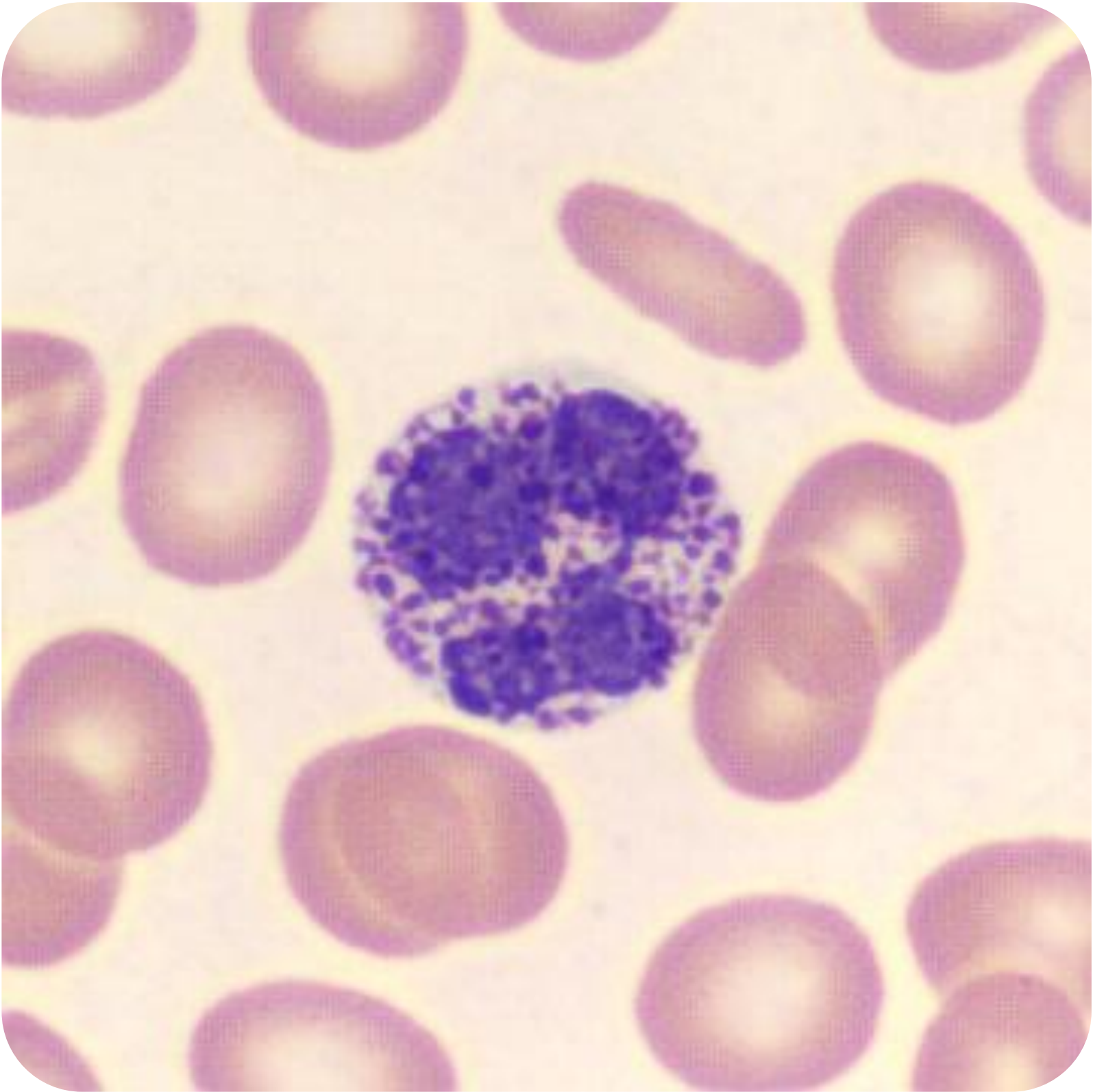
	Conventional Units	SI Units
WBC Total	$3.6 - 10.6 \times 10^3/\mu\text{L}$	$3.6 - 10.6 \times 10^9/\text{L}$
EO, Relative	1 – 3%	0.01 – 0.03
EO, Absolute	$0.0 - 0.3 \times 10^3/\mu\text{L}$	$0.0 - 0.3 \times 10^9/\text{L}$



WBC - Basophil

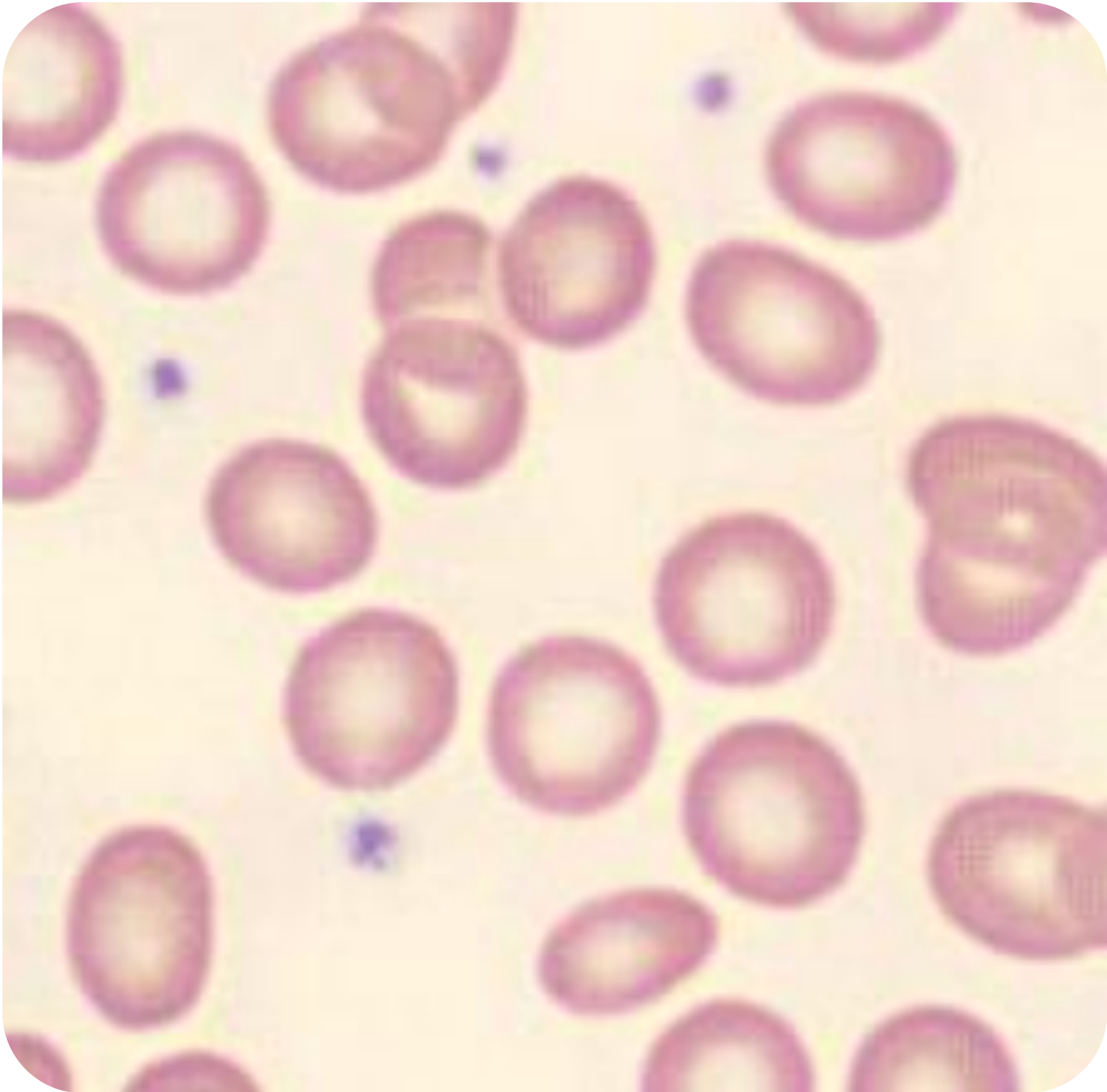
- Role?
- Reference Range?

	Conventional Units	SI Units
WBC Total	$3.6 - 10.6 \times 10^3/\mu\text{L}$	$3.6 - 10.6 \times 10^9/\text{L}$
BA, Relative	0 – 2%	0.00 – 0.02
BA, Absolute	$0.0 - 0.2 \times 10^3/\mu\text{L}$	$0.0 - 0.2 \times 10^9/\text{L}$



RBC and PLT

- Role?
- Reference Range?



	Conventional Units	SI Units
RBC	4.00 – 6.00 x 10 ⁶ /μL	4.00 – 6.00 x 10 ¹² /L
PLT	150 – 450 x 10 ³ /μL	150 – 450 x 10 ⁹ /L